

Machine learning identified molecular fragments responsible for infrared emission features of polycyclic aromatic hydrocarbons

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ABSTRACT

Machine learning feature importance calculations are used to determine the molecular substructures that are responsible for mid- and far-infrared (IR) emission features of neutral polycyclic aromatic hydrocarbons (PAHs). Using the extended-connectivity fingerprint as a descriptor of chemical structure, a random forest model is trained on the spectra of 14 124 PAHs to evaluate the importance of 10 632 molecular fragments for each band within the range of 2.761 to 1172.745 μm . The accuracy of the results is confirmed by comparing them with previously studied unidentified infrared emission (UIE) bands. The results are summarized in two tables available as Supplementary Data, which can be used as a reference for assessing possible UIE carriers. We demonstrate that the tables can be used to explore the relation between the PAH structure and the spectra by discussing about the IR features of nitrogen-containing PAHs and superhydrogenated PAHs.

Key words: astronomical data bases: miscellaneous – software: data analysis – ISM: molecules – infrared: ISM.

1 INTRODUCTION

A series of unique interstellar mid-infrared (IR) spectral emission features are widely detected in the Milky Way and extragalactic galaxies. Their main bands appear around 3.3, 6.2, 7.7, 8.6, 11.2, and 12.7 μm are called unidentified infrared emission (UIE) features. It has been recognized since early years that UIE bands could stem from the vibration of C–H and C–C chemical bonds in organic molecules. This leads to the widely accepted hypothesis that the UIE carriers could be polycyclic aromatic hydrocarbons (PAHs; Leger & Puget 1984; Allamandola, Tielens & Barker 1985). In view of the important role of PAHs in the evolution of interstellar medium (ISM), many efforts are devoted to identifying the PAH molecular structures from UIE bands (Peeters 2011). Despite a few PAHs have likely been detected via radio telescopes in recent years, including PAHs containing C–N bonds (McGuire et al. 2018, 2021; Burkhardt et al. 2021), the identification of interstellar PAHs as UIE carriers still remain challenging, due to the vast variation of the PAH chemical structures and the complex structure–emission relation (Li 2020).

The IR emission features of PAHs are ascribed to characteristic vibrational modes of bonds or functional groups of specific types (Peeters et al. 2021). Traditionally, the 3.3 μm feature is mainly due to C–H stretching, the 6.2 μm feature to C–C stretching, the 7.7 μm feature to coupled C–C stretching and C–H in-plane bending, the 8.6 μm feature to the C–H in-plane bending, and the 10 – 15 μm features to the C–H out-of-plane bending, etc. However, previous studies indicate that observed spectra cannot be adequately explained without considering the specific atomic surroundings of the chemical bond. For instance, the shift of IR peaks around 3.3 μm was not well explained, until the local atomic environment surrounding the C–H

bond is taken into account (Bauschlicher Jr, Peeters & Allamandola 2009). The mixed emission features between 10 and 15 μm is found to rely on the configuration of adjacent aromatic rings carrying different numbers of C–H bonds (Hony et al. 2001). The subsequent investigations concentrated on specific subclass of PAHs as motivated by their potential existence in ISM, accounting for ring defects (Ricca, Bauschlicher Jr & Allamandola 2011; Yu & Nyman 2012; Öttl et al. 2014; Bauschlicher Jr 2015; Devi, Buragohain & Pathak 2020), side groups (Sadjadi, Zhang & Kwok 2015; Bauschlicher Jr 2016; Yang et al. 2017; Buragohain et al. 2020), edges (Sandford, Bernstein & Materese 2013; Candian, Sarre & Tielens 2014; Ricca et al. 2018, 2019; Yang, Li & Glaser 2020), charge (Cecchi-Pestellini et al. 2008; Maragkoudakis, Peeters & Ricca 2020; Devi, Pathak & Vats 2022), size (Draine & Li 2007; Bauschlicher Jr, Peeters & Allamandola 2008; Bauschlicher Jr et al. 2009; Andrews et al. 2015), doping of heteroatoms (Buragohain et al. 2015; Canelo et al. 2018; Oliveira et al. 2021), etc. Despite these efforts, a comprehensive understanding of the correlation between the structure and spectrum of PAHs across the broad IR region is still missing.

The advent of the big data era has brought a new opportunity to improve the situation. Spectra of a large number of organic compounds are made available in data base systems. Remarkably, the NASA Ames PAH IR Spectroscopic Database (PAHdb) has compiled over four thousand spectra of PAHs using ab initio calculations (Bauschlicher Jr et al. 2018) and experiments (Mattioda et al. 2020). This high-quality data has inspired the use of machine learning (ML) methods for a more comprehensive view of the complex spectrum–structure relationship (Calvo et al. 2021; Laurens et al. 2021; McGill et al. 2021). Recently, Kovács et al. (2020) trained a neural network (NN) model for predicting IR spectra of PAHs directly from their chemical structures, and demonstrated excellent predictive skill of the ML model for out-of-sample inputs. In this study, we built an ML

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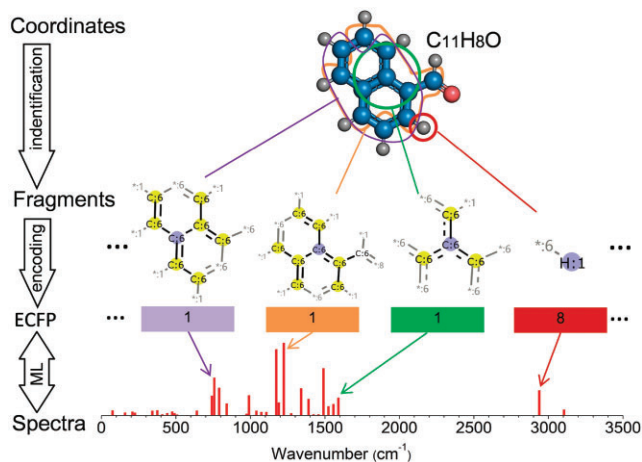


Figure 1. Schematic of the processes for generating the ECFP of 1-Naphthaldehyde. Arrows in colours indicate the correlation between molecular fragments and IR bands. The purple circle corresponds to the fragment centre, the yellow circle represents aromatic atoms, and the asterisk symbol represents adjacent atoms. The atomic number is indicated by the digit following the colon. The grey chemical bonds depict the adjacent portion of the fragments.

model to explore the complex relation between the subtle molecular structure of PAHs and their IR emission features. The molecular structural origins have been identified for individual bands in a wide spectral region.

2 METHODS

2.1 Input: molecular descriptor

The ML prediction of physical properties of a molecule strongly depends on how the molecular chemical structure is mathematically described. Traditionally, molecular geometry in chemical spectra data bases is represented using Cartesian coordinates. However, Cartesian coordinates are not a suitable molecular descriptor for ML since they are not invariant to translation, rotation, and permutation of atom numbers. Therefore, in ML, the Cartesian coordinates are often transformed into an invariant molecular descriptor such as the extended-connectivity fingerprint (ECFP; Rogers & Mathew 2010), the distance or Coulomb matrix (Ivanciuc 2000; Rupp et al. 2012), or so on. In particular, the ECFP shows superior predictive power for the ML prediction of PAH IR spectra compared to other invariant descriptors (Meng et al. 2021).

The ECFP encodes the local neighbourhoods and bonding connectivity of each atom in a molecule into a series of digital identifiers, each of which represents an individual substructure of the molecule, namely a molecular fragment. To identify possible fragments in a molecule, we used the algorithm of Morgan (1965) that systematically records the neighbourhood of each non-hydrogen atom into multiple circular layers up to a cutoff radius (R_{cut}), as implemented in RDKit.¹ Thereby, an integer identifier (a digit) is created for each molecular fragment to count how many times it repeatedly appears at different locations in the molecule. For example, as shown in Fig. 1, eight C–H bonds can be found in $\text{C}_{11}\text{H}_8\text{O}$. So, the number 8 is assigned to the identifier of C–H in the ECFP of $\text{C}_{11}\text{H}_8\text{O}$. Every digit like this representing a specific fragment, is called an *ECFP*

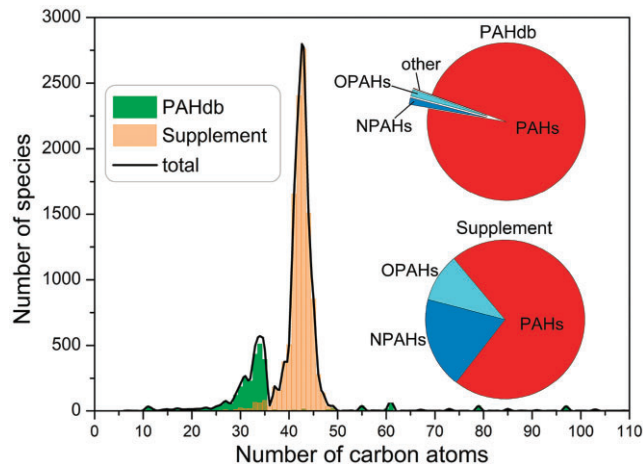


Figure 2. Comparison of PAH size distributions in the two data sets (green for PAHdb, orange for Supplement). Insets show the proportion of PAHs of different types: ‘PAHs’ stands for molecules with solely C and H, ‘NPAHs’ and ‘OPAHS’ for N- and O- containing ones, respectively.

feature (to distinguish from the term ‘emission feature’ for the IR spectrum) in ML. The ECFP is finally created by collecting all the ECFP features into a digital string. The selection of the R_{cut} cutoff radius is a compromise between predictive accuracy and the size of the fragment library, since the prediction error reduces and the fragment number increases with increasing R_{cut} (see the section 1 of the Supplementary Information). Using this approach, 10 632 molecular fragments were in total identified for the PAH molecules considered in this work with $R_{\text{cut}} = 3$.

2.2 Output: IR spectra

The ECFPs and the IR spectra of 14 124 PAHs were used as the input and the output for training the ML model, respectively. The output data set includes the theoretical spectral diagrams of 2925 neutral PAHs from the 3.2 version of the PAHdb², after excluding 1213 charged PAHs and 95 ones whose Cartesian coordinates cannot be successfully transformed into ECFPs by RDKit. We here focused exclusively on neutral PAHs because the topological descriptor we used was unable to distinguish effectively between neutral PAHs and ions, despite the availability of spectra for charged PAHs in the PAHdb data set. As a result, our model was not very accurate in predicting ions, despite our attempts to modify the ECFP. The majority of PAHs in PAHdb contains less than 35 C (see Fig. 2). However, previous studies indicate that UIE bands could more possibly come from larger PAHs (e.g. with ~ 50 C; Tielens 2008).

To complement PAHdb, we have computed the IR spectra of 11 199 PAHs with larger size as an additional data set (namely Supplement, see distribution in Fig. 2). A significant distinction between the molecules present in PAHdb and the supplementary data set lies in their disparate size distributions, resulting in variations in the considered fragments. For example, PAHdb yielded 6468 fragments in total (with $R_{\text{cut}} = 3$), whereas the inclusion of the supplementary data set resulted in the discovery of 10 632 fragments. These spectra have been computed using a Gaussian 16 B.01 (Frisch et al. 2016) implementation of density functional theory (DFT) at the B3LYP/6-31G level with a scaling factor of 0.961, one of the

¹<http://www.rdkit.org/>

²<https://www.astrochemistry.org/pahdb/>

approaches used by PAHdb (Boersma et al. 2014; Bauschlicher Jr et al. 2018).

It should be noted that this work includes a selection of unsaturated PAHs to ensure the representation of the diverse range of potential UIE carriers, taking into account the presence of radicals in the interstellar molecules that have been detected previously (Guélin & Cernicharo 2022; McGuire 2022). The Supplementary Data III provides the chemical formulas, number of unpaired valence electrons, spin multiplicities, simplified molecular input line entry system (SMILES) strings, and xyz coordinates of the PAHs that carry the additional spectra. We also note that in this study, the data set was computed using the harmonic assumption same as PAHdb, which was necessary due to the large size of the systems, even though anharmonic calculations can provide better agreement with experimental data (Mackie et al. 2015, 2016; Lemmens et al. 2019).

The discrete IR intensity is converted to a histogram with a bin width of 16.65 cm^{-1} , determined through the utilization of Knuth's Bayesian rule as implemented in the ASTROPY package (Knuth 2006). This results in 171 bands in the spectral region between 2.761 and $1172.745 \text{ }\mu\text{m}$ (8.527 – 3620.590 cm^{-1}). Note, that there are 54 intervals with no band at all, most of which are in the ranges of 3.720 – $5.030 \text{ }\mu\text{m}$ (1989.340 – 2688.450 cm^{-1}) and of 2.761 – $3.120 \text{ }\mu\text{m}$ (3204.450 – 3620.590 cm^{-1}). Bands beyond $2.761 \text{ }\mu\text{m}$ (3620.590 cm^{-1}) were discarded as there is only a single compound in the whole data base contributing to this region.

Using the method introduced in the previous subsection, we identified 10632 ECFP fragments from the molecular structures of the 14124 PAH molecules, which were used to compose the ECFPs as the input for training the ML model. The input and output data sets, along with the code of ML programme, are provided as Supplementary Data IV for the ease of the readers who are interested in repeating the calculations.

2.3 ML algorithm

The random forests (RFs) algorithm was used to train the model as implemented in the open-source SCIKIT-LEARN library (Pedregosa et al. 2011). The RF algorithm is chosen because it has a built-in metric for the feature importance that is practical for the task of this work, in addition to its efficiency and natural resistance to overfitting (Svetnik et al. 2003). An RF is a combination of decision trees T , every of which comprises a number of nodes t connected through splitting branches. The hyperparameters (such as the number of trees, etc) were previously adjusted for predicting PAH IR spectra (Kovács et al. 2020; Meng et al. 2021), as provided within the ML code in Supplementary Data IV.

In Section 2 of the Supplementary Information, we conducted a test to assess the impact of changes in hyperparameters and data sets on our model outputs. These results indicate that our model remains resilient to such variations. Details regarding the accuracy of the model, which includes the outcomes of a 5-fold cross-validation, are available in Section 3 of the Supplementary Information. These results demonstrate that our models exhibit strong out-of-sample predictability.

For each of the 171 frequency intervals, we trained a RF model to assess the importance of each ECFP feature on a specific IR emission feature, since an IR emission band could strongly be related to a specific molecular fragment that are represented by an ECFP feature X_m (e.g. indicated by the arrows in Fig. 1). In ML analysis, this correlation is evaluated by the importance of X_m for predicting the spectrum, namely the feature importance $FI(X_m)$. It can be calculated by adding up the weighted impurity decreases $p(t)\Delta i(s_t, t)$ for all

nodes where X_m is used, averaging over all N_T trees in the forest,

$$FI(X_m) = \frac{1}{N_T} \sum_T \sum_{t \in T: v(s_t)=X_m} p(t)\Delta i(s_t, t), \quad (1)$$

where $p(t)$ is the proportion N_t/N of samples reaching t , $v(s_t)$ is the variable used in the split s_t . The Gini index is used as the impurity function $i(t)$ (Loupe et al. 2013). The sum of the importance of all ECFP features is normalized to 1.0. As 10632 ECFP features are included in this work, the average value of $FI(X_m)$ will be as low as 9.4×10^{-5} . An FI at the order of $>10^{-3}$ or 10^{-2} could therefore be prominent over a band. We note that there is no definite correlation between FI and the band's intensity. The RF algorithm incorporates a random component that may introduce uncertainty (noise) into the outcomes. After conducting a preliminary assessment, it has been determined that the average random noise in our feature importance results is estimated to be around 3.58 per cent of $FI(X_m)$.

3 RESULTS AND DISCUSSION

The importance of an ECFP feature, denoted as FI , indicates the level of importance of a molecular fragment in relation to a given emission band. By evaluating FI , we have identified the molecular fragments responsible for each band between 2.761 and $1172.745 \text{ }\mu\text{m}$. The findings are summarized in two oversized tables, which are included as the Supplementary Data I and II of this article. The first table lists the top 10 molecular fragments in descending order of importance for each emission band, while the second table includes the top 100. These tables offer a valuable tool for assessing potential carriers of UIE. It is important to note that while FI aids in identifying the molecular structure class associated with a specific band, there is no definitive correlation between FI and the absolute band intensity. Considering the extensive amount of information encompassed by these tables, we have carefully selected a few emission features that are of significant astronomical interest for detailed discussion in the subsequent text.

To validate our model, the results are first compared with the established sources of extensively studied UIE bands. These include the 3.3 , 7.7 , 8.6 , 11.2 , and $12.0 \text{ }\mu\text{m}$ bands that are known to arise from C–H stretching, C–C stretching, C–H in-plane bending, and C–H out-of-plane bending vibration, respectively. It is important to note that the results presented for the 7.7 , 8.6 , and $12.0 \text{ }\mu\text{m}$ bands in this study pertain to neutral PAHs, despite these bands being predominantly associated with strong cationic components (Bauschlicher Jr et al. 2009). The ML results shown in Fig. 3 are found to be consistent with the classical theory. Interestingly, the methyl group is found to make a noticeable contribution to the band at $3.3 \text{ }\mu\text{m}$ (as highlighted by the orange ellipses in Fig. 3), in addition to its well-known contribution to the $3.4 \text{ }\mu\text{m}$ band (Steglich et al. 2013). Contrary to the results of Candian et al. (2012) suggesting that UIE bands slightly blue-shifted from $3.3 \text{ }\mu\text{m}$ may originate from PAH with bay-like structures, the molecular fragments with bay-like structures were not found to dominate the bands at $3.29 \text{ }\mu\text{m}$.

Regarding the $7.7 \text{ }\mu\text{m}$ band, the fragments responsible, as illustrated in Fig. 3(b), suggest that C–C bonds within the carbon skeleton hold greater significance compared to those located at the edges. As shown in Fig. 3(d), it is seen that the $11.2 \text{ }\mu\text{m}$ band is primarily characterized by solo C–H bonds (without contiguous H atoms), aligning with the out-of-plane bending vibration mode of neutral PAHs previously investigated by Maragkoudakis et al. (2020). In the case of the $12.0 \text{ }\mu\text{m}$ band, the C–H bonds of the trio type (with three contiguous H atoms) are found to be equally prominent as the

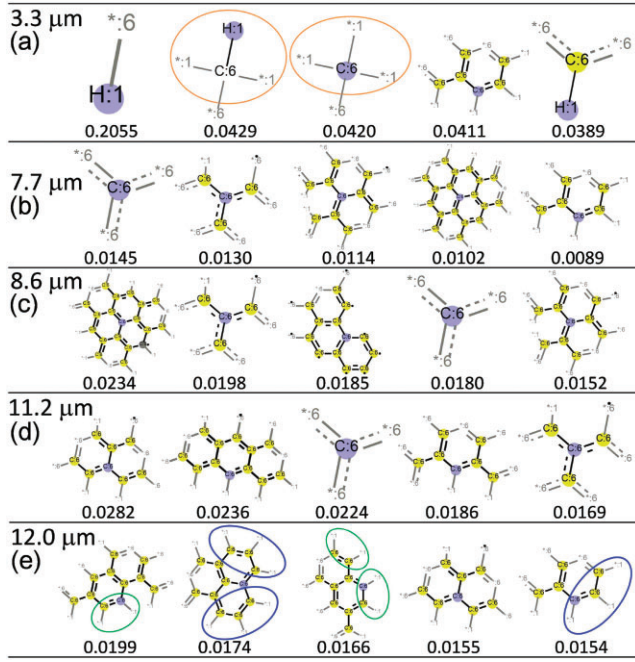


Figure 3. Five most important fragments for the bands at 3.3, 7.7, 8.6, 11.2, and 12.0 μm , sorted from left to right with descending FI (printed at the bottom).

commonly considered duo type bonds, as highlighted by the blue and green ellipses in Fig. 3(e). Furthermore, carbon skeleton fragments demonstrate relatively high importance across the entire spectrum of bands. This observation is to be expected, as these fragments are indicative of the characteristic aromatic structure. However, the positioning of these fragments among the top fragments for a given band indicates a reliance on the specific carbon skeleton structure associated with that particular band, as illustrated in the Section 4 of the Supplementary Information. It is noteworthy that a wavelength of 1172.745 μm is present, at which emission bands are exhibited by 904 PAHs in our data set. As demonstrated in Section 5 of the Supplementary Information, a visualization of a sample molecule's vibration suggests that this feature is likely correlated with the out-of-plane twisting of the molecule.

Statistically, when a band is dominated by the vibration characteristics of a particular molecular fragment, the ECFP feature corresponding to which will exhibit an importance significantly higher than that of other fragments. Consequently, the distribution of FI of a distinguishing band should be non-uniform, and may represent a particular structural class of PAHs. To visualize the FI distribution for different bands, we used a colour scale in Fig. 4(a). Notably, several IR bands exhibit a non-uniform colour scale, suggesting that they may be distinguished by the dominance of some specific molecular fragments.

To display the non-uniform distributions more clearly, we plot the sum of FI of the five most important fragments for different bands in Fig. 4(b). Roughly, the sum is supposed to be high for a distinguishing band. The blue bars in this figure highlight a number of prominent bands that are intensively discussed in the literature, such as the aforementioned ones at 3.3, 6.2, 7.7, 8.6, 11 – 15 μm . The 3.21, 3.27, 5.1, 7.0, 11.9, 13.2 μm bands (green bars) are also noticeable in UIE (Bauschlicher Jr et al. 2009; Bauschlicher Jr & Ricca 2013; Sundararajan et al. 2018; Ricca et al. 2019). Our findings also identify several unique bands (indicated by red bars) that may be of interest

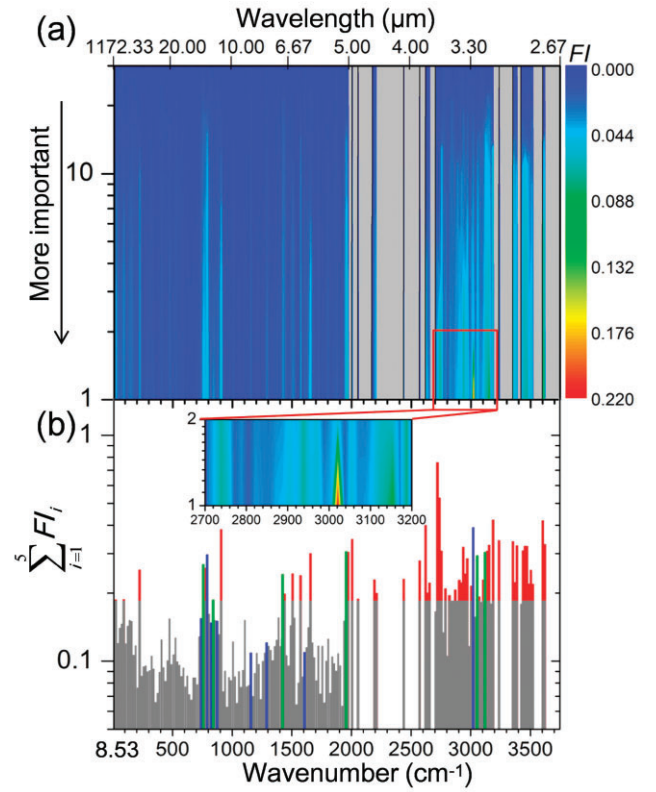


Figure 4. (a) FI distribution for the bands between 2.67 and 1172.33 μm . Upwards along the vertical axis, the importance of 100 ECFP features are sorted from high to low, and is represented by the colour scale. The regions with no spectrum are shown in grey. (b) Sum of FI of the five most important fragments for different bands. The illustration is a zoom of panel (a) at a range of about 3.3 μm (3000 cm^{-1}). The FI maximum shown for this band is associated with fragment 1 in Fig. 3(a).

for future astronomical observations of PAHs (with FI sum above the mean of the blue bars). The wavelengths associated with these bands are 44.46, 12.92, 11.02, 6.64, 6.36, 6.04, 5.07, 4.99, 4.98, 4.86, 4.1, 4.56, 4.52, 4.1, and 3.89 – 2.76 μm .

Among the distinguishing emission features shown in Fig. 5, the 2.9, 3.1, and 3.7 μm ones are characteristic for nitrogen-containing PAHs (NPAHs). NPAHs are of importance for astronomical observations aiming at the abiogenesis question as they could be building blocks of biomolecules. Previous studies indicate that the stretching of N–H bonds gives rise to emission bands at 2.8 – 3.1 μm (Vats et al. 2022), while bands around 4.5 μm are suspected to stem from the stretching of $\text{C}\equiv\text{N}$ bonds (Allamandola et al. 2021). Here, further structural dependence of these emission features is revealed by ML. Our results indicate that the bands at different frequencies in the spectral region between 2.5 and 6.5 μm is respectively dominated by NPAHs of different structural classes, namely amine ($-\text{NH}_2$), imine ($-\text{NH}$), cyanide ($-\text{CN}$), N embedded inside (N_{endo}), and on edges (N_{exo}), as shown in Fig. 5. We see that the bands at 2.761/2.867, 3.136, and 4.523/4.559 μm are dominated by the fragments of the $-\text{NH}_2$, $-\text{NH}$, and $-\text{CN}$ types, respectively. This is in line with previous studies that have attributed the 2.8 – 3.1 μm bands to the stretching of N–H bonds and the 4.5 μm bands to the stretching of $\text{C}\equiv\text{N}$ bonds. For instance, Rozenberg, Fausto & Reva (2021) reported that the 2.801 and 2.897 μm bands correspond to the antisymmetric and symmetric stretching of N–H bonds in $-\text{NH}_2$, respectively, while Góbi et al. (2019) suggested that the 3.1 μm band is due to the stretching of the

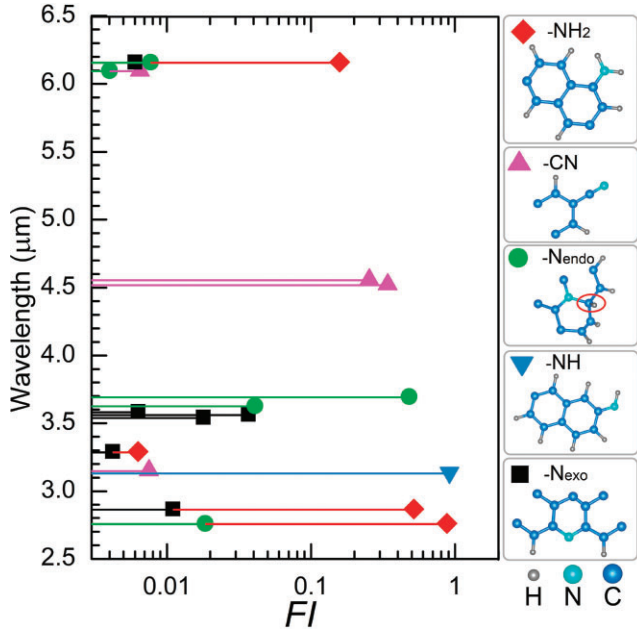


Figure 5. Left: FI of fragments in different classes for the bands between 2.5 and 6.5 μm. Right: Example fragments for five NPAH structural classes.

N–H bond in –NH. Our results also identify other emission features that could be attributed to NPAHs of different structural types, such as amine, imine, and cyanide, highlighting their potential significance for future astronomical observations.

Specifically, our results suggest that NPAHs containing N_{endo} and –NH₂ fragments are characterized by distinctive emission peaks at 3.674 and 6.160 μm, respectively. The blue shift of the band at 6.2 μm (known as the class A band) has been previously proposed as a tracer for interstellar NPAHs, with N_{endo}-type fragments responsible for the blue shifted emission (Hudgins, Bauschlicher Jr & Allamandola 2005; Ricca, Boersma & Peeters 2021). However, our results indicate that the blue shifted emission at 6.16 μm is mainly attributed to –NH₂-type fragments. Fig. 5 reveals that fragments with the –NH₂ sidegroup are ten times more important than N_{endo} or N_{exo}-type fragments in contributing to this emission feature. Additionally, we observe that the emission at 3.674 μm is highly correlated with N_{endo} fragments, specifically those with *sp*³-hybridized C, such as the one marked with a red circle in the right-hand panel of Fig. 5. These results shed new light on the structural dependence of NPAH emission features.

Accompanying the dominant 3.3 μm emission feature, emissions near 3.4 μm are widely detected towards interstellar regions (Sloan et al. 1997). Their origin is under debate between several paradigms, including bond anharmonicity (Barker, Allamandola & Tielens 1987), aliphatic side-groups (Joblin et al. 1996) and superhydrogenation (SH; Schutte, Tielens & Allamandola 1993). Recently, Kwok (2022) shows that a number of observations contradict the anharmonicity explanation (Goto et al. 2003; Van Dienenhoven et al. 2004). Also, Sandford (1991) found that it is hard to fit the relative intensity of the bands at 3.40, 3.46, 3.51, and 3.56 μm towards NGC 7027 using the spectra of PAHs with aliphatic side-groups. On the other hand, evidences have emerged in recent years supporting the SH origin of IR spectra (Materese, Bregman & Sandford 2017; Mackie et al. 2018; Yang et al. 2020). However, it remains challenging to systematically match the relative intensity of the emissions around in the 3.4 μm range.

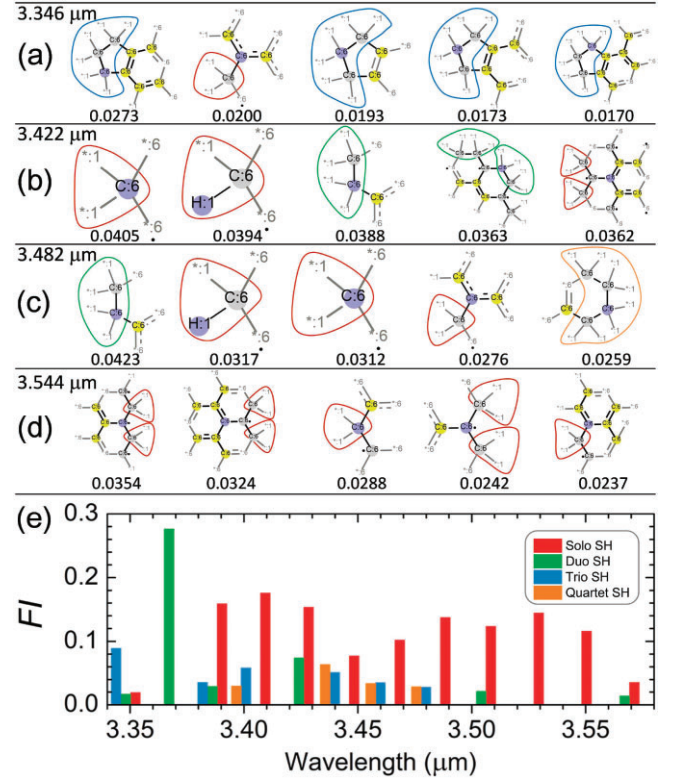


Figure 6. Five most important fragments for the bands at 3.346 (a), 3.422 (b), 3.482 (c), and 3.544 μm (d), sorted from left to right with descending importance. (e) Distribution of the FI of different SH substructure classes for the bands near 3.4 μm.

Moreover, our ML results reveal a dependence of the emission features around 3.4 μm on different substructures of SH PAHs. As shown in Fig. 6(a)–(d), the bands at 3.346, 3.422, 3.482, and 3.544 μm are attributed to SH PAH substructures of four different classes. These classes include SH C sites without adjacent SH atoms (referred to as ‘Solo SH’, illustrated by red coils), those with two adjacent SH atoms (termed ‘Duo SH’, represented by green coils), those with three adjacent SH atoms (labelled ‘Trio SH’, depicted by blue coils), and those with four adjacent SH atoms (called ‘Quartet SH’, illustrated by orange coils). Fig. 6(e) shows that the emission of Solo SH PAHs is red shifted in comparison to that from SH PAHs belonging to the other three classes. Furthermore, in the 3.523 μm bands, a fragment referred to as ‘tertiary C’ in prior studies (Pla et al. 2020) has been identified. However, its significance is slightly lower compared to superhydrogenated molecular fragments.

4 CONCLUSIONS

The analysis of ML feature importance was employed to determine the structural classes of PAHs that could potentially account for IR emission bands across a broad spectral range. The resultant tables, accessible as Supplementary Data in this article, establish a connection between the chemical structures of 14 124 neutral PAHs and their corresponding IR bands. To validate the findings, several well-known UIE bands were utilized as benchmarks. Furthermore, numerous distinctive bands were identified, which hold potential significance for astronomical observations of PAHs, many of which fall within shorter wavelengths compared to the commonly examined bands in the PAH ‘fingerprint’ region.

To showcase the potential utility of the Supplementary Data in exploring the intricate relationship between PAH structures and spectra, our results were applied to investigate the origin of emission features in NPAHs and SH PAHs. The obtained results largely align with previous studies but provide additional insights into the structural dependency of bands within the 2.7 – 6.2 μm range. Beyond the commonly studied molecular substructure types, we have identified distinct molecular substructure types through analysing the statistics of important fragments across all bands. These include substructures with topological defects (such as C4, C5, and C7 rings), Si substitution, and hydroxyl groups. Furthermore, the substructures characterized by zigzag, armchair, or gulf edges hold potential significance in unraveling the formation processes of PAHs (Qi et al. 2018; Hanine et al. 2020). We highlight these structures as suggestions for further investigation.

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DATA AVAILABILITY

Supplementary Data I and II contain extensive tables that lists the top 10 and the top 100 molecular fragments responsible for the spectral bands between 2.761 and 1172.745 μm , respectively. Supplementary Data III provides the chemical formulas, number of unpaired valence electrons, spin multiplicities, SMILES strings, and xyz coordinates of the PAHs that carry the additional spectra. Supplementary Data IV includes the ML training code. Supplementary Information contains validation results of the ML model. Supplementary Data are available on Zenodo at DOI:10.5281/zenodo.7758208.

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Supplementary data are available at *MNRAS* online.

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